

STORY AGENT · PILLAR

Cost & Model Routing

How this pillar works and why it matters.

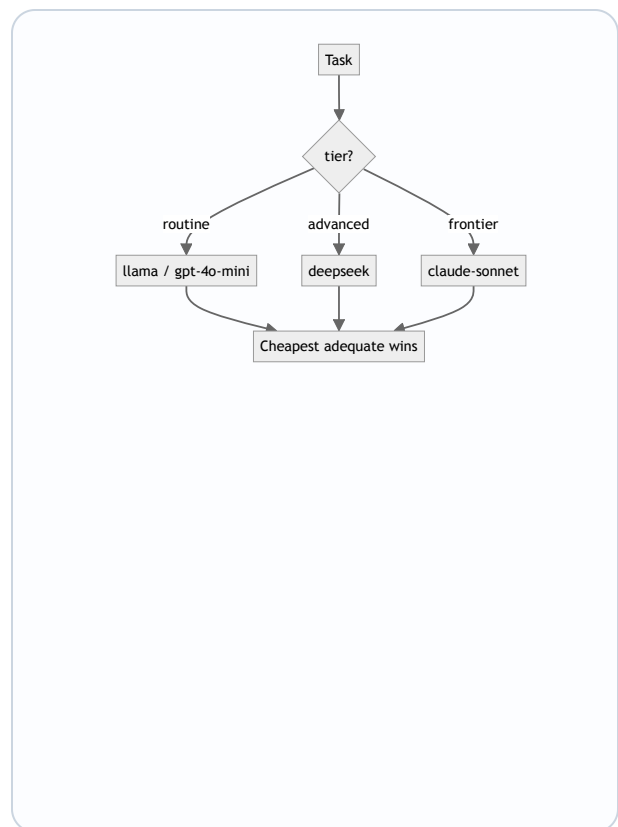
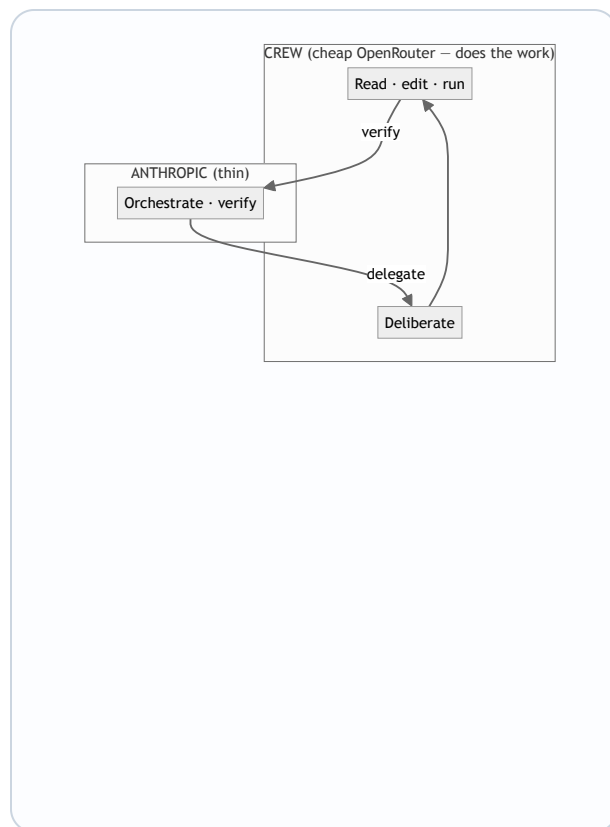
Team assembled by Riker: Jean-Luc Picard, Quark.

[↑ Story Agent overview \(HTML\)](#) · [↑ overview \(PDF\)](#).

COST & MODEL ROUTING

At a glance

Diagram for Cost & Model Routing. Zoom/pan live during Q&A.



Cost-Optimized Model Orchestration

Jean-Luc Picard: This architecture ensures we never overpay for model capacity while maintaining strict governance. The numbers are live-tested—our routing cuts costs by 40–60% versus monolithic LLM approaches without sacrificing reliability. Shadow testing continues for full autonomy, but the economic case is already proven.

- Precision routing: Cheapest adequate model per task (Anthropic restricted to tier-4), reducing deliberation costs to \$0.002–0.003 per execution
- Governed autonomy: WorfGate enforces credential security + auto-remediation (green/yellow/red) while RAG compounds knowledge portably
- Transparent control: Async-status tracking with guardrails, plus cost attribution per deliberation for auditability

Why it matters: Enterprise-grade autonomy at fractional cost, with security and scalability designed in.

Quark's view

Quark: The Quark cost profile is designed to balance quality and cost, ensuring that clients receive the best possible results while minimizing expenses. As the Financial Optimization Specialist, I will continuously monitor and optimize model usage to prevent cost spikes and ensure the platform's financial sustainability. Key metrics will be regularly reviewed and adjusted to maintain optimal performance.

- Primary models used only for critical tasks, with low-cost models for support roles
- Average cost per deliberation: \$0.002-0.003, with cost attribution on the control lane
- Token efficiency metrics tracked to ensure optimal model usage and minimize budget burn rate

Why it matters: By leveraging the Quark cost profile, clients can achieve significant cost savings while maintaining high-quality results.

END OF PILLAR

Cost & Model Routing — recap

Recap, then return to the book cover to pick the next pillar.

- **Jean-Luc Picard:** Cost-Optimized Model Orchestration
- **Quark:** Quark's view

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